Quantitative Colocalization Analysis of CCT1 and Mutant Huntingtin Aggregates in Fluorescence Microscopy Images

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I. SUMMARY

This report presents an automated pipeline to quantify and map colocalization between the molecular chaperone CCT1 and mutant Huntingtin (mHTT) aggregates in 3D fluorescence microscopy images. Robust spatial quantification of CCT1 at mHTT inclusions has previously been lacking, limiting

systematic analysis of CCT1–mHTT colocalization. To address this, a transparent, GUI-driven pipeline was developed that combines statistical segmentation with intensity-based enrichment analysis. The pipeline processes multi-channel z-stack images, segments mHTT aggregates using voxel-based masks, and quantifies the spatial relationship between CCT1 and mHTT through complementary colocalization and enrichment metrics. Validation was performed on expert-annotated microscopy data, with visual overlays ensuring biological plausibility of automated outputs. Results suggest a significant enrichment of CCT1 at mHTT aggregation sites in the analyzed dataset. The pipeline produces reproducible and interpretable outputs, providing a flexible and user-friendly alternative to previously manual, labor-intensive, and subjective workflows. Its modular architecture supports integration of future AI models and enables scalable, high-throughput analysis of CCT1–mHTT interactions in evolving microscopy datasets.

II. Introduction

A. Context

Huntington's disease (HD) is a fatal neurodegenerative disorder caused by a mutation in the **HTT** gene, leading to progressive motor, cognitive, and psychiatric symptoms [1]. It follows an autosomal dominant inheritance pattern, meaning that a child has a 50% chance of inheriting the disease if one parent carries the mutated gene [2]. The mutation results from an abnormal expansion of CAG trinucleotide repeats [3], which are three-letter DNA sequences (cytosine-adenine-guanine) that code for the amino acid glutamine. This leads to a longer polyglutamine (polyQ) tract in the huntingtin (HTT) protein. Healthy people typically have fewer than 35 CAG repeats, while those with 36 or more develop HD, with longer repeat expansions leading to an earlier onset and more severe progression of the disease [4].

1) Genetic and Molecular Basis of Huntingtin: Huntingtin is essential for neuronal function, but the expanded polyQ tract in mutant huntingtin (mHTT) disrupts its structure. This change in shape makes it more likely to misfold. [3]. Misfolded proteins fail to function properly and tend to stick together, forming **aggregates**. Aggregates are clumps of misfolded proteins that accumulate inside cells. These mHTT aggregates, which appear in both the nucleus (the control center of the cell) and cytoplasm (the fluid-filled space outside the nucleus), disrupt normal cell functions, interfering with genetic regulation, energy production in mitochondria, and communication between neurons, ultimately leading to cell death and neurodegeneration [5], [3].

2) Role of Molecular Chaperones: Cells have defense mechanisms to counteract the effects of misfolded proteins. One key mechanism is the use of molecular chaperones, which are specialized proteins that assist in the folding and refolding of other proteins to maintain cellular health [6].

Among these, **CCT1** is of importance due to its interaction with mHTT. Studies have shown that increasing CCT1 levels in cells leads to a reduction in mHTT aggregates and improved neuronal survival, suggesting that CCT1 could be a potential therapeutic target for HD treatment [3]. However, the exact mechanisms by which CCT1 recognizes and interacts with mHTT aggregates remain poorly understood [4]. One approach to gaining insight into this interaction is to examine whether CCT1 physically associates with mHTT aggregates within cells, observed through colocalization.

3) Colocalization of CCT1 and mHTT Aggregates: Colocalization refers to the spatial overlap of two molecules within a cell, suggesting that they physically interact or function in the same cellular pathways. Foundational colocalization studies have shown that CCT1 is often found at the same locations as mHTT aggregates, suggesting an attempt by the chaperonin system to counteract mHTT toxicity [4]. In this project, fluorescence microscopy data will be used (see: Fig. 4). Fluorescence microscopy is a widely used technique for visualizing these molecules within cells by labeling them with fluorescent markers; this labeling process is called staining. Staining allows researchers to track their locations and potential interactions.

Understanding the extent and nature of the interaction between CCT1 and mHTT is critical for future research. If CCT1 is actively involved in breaking down or neutralizing toxic mHTT aggregates, then enhancing its function could be a promising therapeutic strategy.

Problem

While studies suggest that CCT1 plays a role in regulating mutant huntingtin aggregation [4], Quantifying CCT1's spatial relationship within huntingtin cells seems to remain a challenge, leading to a situation where the extent and nature of this spatial relationship remain poorly quantified. Traditional fluorescence microscopy-based colocalization analysis relies on manual methods which are [7]:

- Time-consuming and labor-intensive
- Subject to human bias
- Limited in reproducibility across experiments

The system currently being used to determine colocalization is called Fiji [8]. Fiji is an open-source image processing package built on ImageJ [9], tailored specifically for biological-image analysis [8]. It integrates a curated collection of plugins and tools, facilitating image processing tasks. Fiji enables the researcher to perform prototyping of image-processing. Because it is a closed system it cannot be combined with additional (python based) approaches. Fiji has been widely adopted for tasks such as segmentation, registration, and visualization of multidimensional biological images, but is subject to the issues mentioned above.

Research question

Does the molecular chaperone CCT1 show significant spatial enrichment at mutant Huntingtin (mHTT) aggregate sites in human cells, and can this relationship be robustly quantified using automated analysis?

B. Existing work

Pixel-intensity correlation methods have been used to measure the degree of overlap between two fluorescent signals. The most common of which are Pearson's correlation coefficient (PCC), which assesses the linear correlation of intensity values in two image channels [10] [11], and Meander Correlation coefficient [11] which refines traditional correlation by taking into account the spatial continuity and local intensity fluctuations along morphological structures. These metrics are somewhat of a standard for colocalization analyses in biology [12]. In practice (as when using FIJI [8]) images are often thresholded to exclude background before computing PCC or Manders' coefficients [13]. Comprehensive guides in the literature outline best practices for these calculations, emphasizing careful background subtraction, appropriate sampling, and the complementary use of scatterplots to visually inspect pixel correlations. These foundational techniques (see: III-A) allow quantification of colocalization in CCT1–mHTT images.

Alongside pixel-wise metrics, object-based colocalization methods have been implemented. Object-based approaches first segment discrete structures (e.g. protein aggregates or organelles) in each channel, then evaluate co-occurrence or spatial proximity of these objects [14]. It checks how many mHTT aggregate objects have a chaperone signal object at the same location. Some recent software tools such as SpotitPy (2022) already implement semi-automated object segmentation and colocalization quantification in high-throughput images [14]. SpotitPy allows robust segmentation of fluorescent puncta (small, bright spots in microscopy images that indicate localized concentrations of biomolecules) and computes the fraction of objects that overlap between channels, enabling cell-by-cell analysis of colocalization events in large datasets. Similarly, ColocalizR (2019) provides an open-source pipeline for high-throughput colocalization analysis on a per-cell basis, integrating cell segmentation with calculation of overlap metrics(PCC and MOC) within the segmented regions, therefore they are somewhat of hybrid between pixel-wise and object-based colocalization [15]. After testing both of these, they were found to be inadequate due to their lack of modularity and 3D capabilities. Instead, SpotitPy and ColocalizR have been analyzed as inspiration for implementing object-based methods. Object-based methods can complement traditional PCC/Manders analyses by accounting for spatial relationships and complex patterns (for instance, distinguishing one large aggregate co-locating with dispersed chaperone signal versus concentrated overlap). In this context, such approaches are useful because mHTT often forms discrete inclusions, while CCT1 displays a predominantly diffuse signal with areas of local enrichment. Segmentation-based analysis could quantify how many CCT1 puncta reside on or near mHTT inclusion bodies.

Colocalization studies using fluorescence microscopy have shown that CCT1 is often found at the same locations as mHTT aggregates, indicating an attempt by the chaperonin system to counteract mHTT toxicity [4]. A study by Sontag et. al. showed that delivery of CCT1 into cells reduced mHTT aggregation and toxicity, suggesting a direct and functional interaction [3].

While some discrepancies exist, such as minimal effects in certain in vivo models [4], the overall consensus is that CCT1 physically interacts with and modulates mHTT aggregation. Prior studies have therefore motivated further investigation of this interaction.

Tools like AggreCount (2020) are already used, providing an automated ImageJ-based macro to identify and count aggregates in cells [7]. AggreCount can classify aggregates as cytosolic or nuclear and output per-cell aggregate counts and areas. It exemplifies the move toward automation, but relies on simple 2D watershed segmentation, which fails for cells with intricate shapes like neurons.

In summary, the existing literature spans from fundamental colocalization metrics (Pearson, Manders, etc.) to advanced imaging analyses relevant to our project. Over the past 5–10 years, the CCT1-mHTT relationship, and related phenomenon has been explored with increasing technical sophistication, including high-resolution structural methods [16] and high-throughput imaging with automated analysis [14] [15].

C. GAP

The concept for this research arose from a convergence of two unmet needs in neurodegenerative disease research: the lack of quantitative, 3D analyzed data on molecular chaperone activity in relation to protein aggregation, and the limitations of current image analysis tools in capturing such interactions reliably at scale.

Past observations from fluorescence microscopy suggested colocalization between CCT1 and mutant huntingtin (mHTT) aggregates [4], but the absence of automated, reproducible analysis pipelines has constrained deeper exploration of this relationship. In this context, the automated pipeline developed in this report aims to rigorously quantify the spatial colocalization between CCT1 and mHTT aggregates, providing a foundation for future biological studies to explore the functional implications of this association

D. Proposal

Rather than relying on pixel intensity alone, the proposed pipeline integrates biological structure, spatial context, and segmentation-derived insights. This allows for more rigorous and potentially biologically meaningful interpretations of colocalization

The unique value of this work lies in its ability to bridge computational rigor and biological relevance. It moves beyond legacy methods like AggreCount by building a flexible and scalable platform that aligns with modern deep learning practices. Not only does this create an easy-to-use, scalable, and reproducible way of performing colocalization analysis, it also reduces bias by formalizing key steps such as thresholding and segmentation within a transparent pipeline. This ensures that results are consistent across runs and that any manual

adjustments made by the user are explicitly controlled and documented.

Additionally, by making the pipeline modular and GUI-accessible, this project empowers broader adoption among non-programming researchers, accelerating translational insights in HD research and beyond.

The value proposition of this project is to provide an automated colocalization analysis pipeline **using** multi-channel fluorescence microscopy images of mHTT and CCT1-stained cells **to** train models for segmentation and generating a colocalization measurement **we can help** biomedical researchers **with a better way to** analyze the 3D spatial relationship between CCT1 and mutant Huntingtin aggregates **so that** they can obtain reproducible, quantitative colocalization metrics **with** automated analysis and visual overlays — without time-consuming manual annotation or overly subjective interpretation.

III. Background

A. Basic knowledge

The quantitative analysis of colocalization between mutant Huntingtin (mHTT) aggregates and the CCT1 chaperone relies on a set of well-established image analysis metrics. These metrics operate on voxel-level data derived from 3D fluorescence microscopy images. Understanding the following key concepts and statistical tools is essential for interpreting the results presented in this thesis. The relevant measures and concepts used in this work are briefly introduced below.

Voxel: In 3D fluorescence microscopy, the basic unit of image data is called a *voxel* (short for "volume pixel"), the three-dimensional equivalent of a pixel [**?**]. It represents a small cuboid volume of the sample, with dimensions defined by the microscope's x, y, and z resolution settings. Each voxel stores an associated fluorescence intensity value for each image channel, enabling quantitative measurements in 3D space. This allows analyses to capture the full spatial structure of biological samples, rather than being limited to individual two-dimensional slices. In this thesis, voxel-based measurements are used throughout the analysis to compute intensity correlations, enrichment levels, and spatial overlap between mHTT aggregates and CCT1 signal in volumetric fluorescence microscopy data.

Pearson's Correlation Coefficient: Pearson's Correlation Coefficient (PCC) is used as a core metric within this study. PCC measures linear correlation between two fluorescence intensities but does not distinguish between co-occurrence and true interaction. A PCC of 1 indicates perfect correlation (colocalization) and 0 indicates random localization.

However, PCC can be sensitive to background and differences in signal dynamic range [10]. To address some of these issues, Manders' overlap coefficients were introduced as an intensity-overlap measure that is independent of signal proportionality.

1) Manders' Coefficients: Manders' Coefficients (M1, M2): Quantify the fraction of one protein's fluorescence overlapping with the other (and vice versa) but do not account for intensity variations, making them prone to overestimation in cases of high background fluorescence.

2) Dice similarity coefficient: The Dice Similarity Coefficient (DSC) quantifies the spatial overlap between two binary sets (e.g. segmentation masks) and is defined as

$$\mathrm{DSC}(A, B) = \frac{2\,|A \cap B|}{|A| + |B|}\,, \tag{1}$$

where $|A|$ and $|B|$ denote the cardinalities of sets A and B, respectively [17], [18].In segmentation terms, (1) can be rewritten as $\mathrm{DSC} = 2TP/(2TP + FP + FN)$, highlighting that DSC is the harmonic mean of precision and recall [19]. High DSC values are achieved when the number of true positives (TP) is high, and both false positives (FP) and false negatives (FN) are low. Notably, voxels that both masks already agree are background (true negatives, TN) are excluded from the calculation, so the score reflects only how well the masks match on the object region and is not diluted by large background areas [20].

DSC is the fraction of all voxels labelled "object". It ranges from 0 (no overlap) to 1 (perfect overlap).

DSC is the de facto metric for validating automated segmentations of biological structures against expert ground truth in both 2-D and 3-D data [20]. It is reported in major benchmarks such as the 2018 Data Science Bowl nucleus-segmentation challenge [21]. In this study mutant Huntingtin aggregates vs. CCT1 chaperone have been compared with DSC to yield a symmetric, intensity-independent overlap coefficient. Conceptually it is similar to Manders' overlap coefficient but it operates on segmented rather than intensity data [12].

3) Mann-Whitney U Test: The Mann-Whitney U test (also known as the Wilcoxon rank-sum test) is a non-parametric statistical test used to determine whether two independent samples originate from the same distribution [22]. Mann-Whitney U ranks all observations and evaluates if one group tends to yield higher ranks than the other [22], [23]. This feature makes it well-suited for the fluorescence intensity data, which during EDA has been determined to deviate from Gaussian distributions due to heterogeneous pixel values and outliers. It provides a robust way to compare median tendencies or overall distributions between two groups without relying on distributional parameters.

In the context of this thesis, a two-tailed Mann-Whitney U test is applied in the Intensity Enrichment Analysis on a per-image basis to compare CCT1 fluorescence intensities inside versus outside mHTT aggregate regions. Here, binary masks derived from fluorescence microscopy segment the image into two sets of pixels: those within mHTT aggregate areas and those in the surrounding non-aggregate region. By comparing the distribution of CCT1 intensities in these two sets, the test assesses whether there is a statistically significant difference in signal localization (so if there is more CCT1 signal in the mHTT aggregates than outside).

4) Cohen's d : Cohen's d quantifies the standardized difference between two distributions and complements significance tests like the Mann–Whitney U by reporting effect size [24],

[25]. It helps in understanding the magnitude of the difference, rather than just whether the difference is statistically significant (which is what a p-value tells you).

For each image, voxel intensities(as defined in Section III-A) are split into two groups based on the mHTT mask: inside-aggregate and outside-aggregate. Cohen's d is computed as:

$$d = \frac{\mu_{\text{in}} - \mu_{\text{out}}}{\sigma_{\text{pooled}}}, \tag{2}$$

where μ_{in} and μ_{out} are the means, and σ_{pooled} is the pooled standard deviation of the two groups.

While the Mann–Whitney U test evaluates whether the difference is statistically significant, Cohen's d expresses its magnitude in interpretable units. Because this analysis compares individual pixel intensities across large 3D microscopy images, the resulting U-values are correspondingly large, reflecting the high number of pairwise comparisons rather than the strength of the biological effect. This standardization enables meaningful biological interpretation and cross-image comparison of CCT1 enrichment strength.

5) Intersection over Union (IoU): Intersection over Union (IoU), also known as the Jaccard index, measures the spatial overlap between a predicted segmentation and ground truth by computing the ratio of intersection to union of their binary masks [26]. For predicted mask A and ground-truth mask B, it is defined as:

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \tag{3}$$

In 3D segmentation of fluorescence microscopy data, this corresponds to the number of overlapping voxels divided by the total number of voxels marked as object by either mask. IoU can also be written in terms of true positives (TP), false positives (FP), and false negatives (FN): $\text{IoU} = \frac{TP}{TP+FP+FN}$ [20].

Like the Dice Similarity Coefficient (DSC), IoU excludes true negatives and focuses on overlap in object regions. However, IoU is stricter: it penalizes partial matches more than Dice, yielding lower values for the same segmentation mismatch [27]. The metrics are mathematically related by $\text{DSC} = \frac{2\,\text{IoU}}{1+\text{IoU}}$ [28], ensuring consistent ranking despite differing scales.

IoU is widely reported in biomedical image segmentation alongside DSC [20]. While DSC is often the primary benchmark, IoU adds value as a more conservative measure of segmentation accuracy, especially in evaluating precise boundary agreement in 3D fluorescence datasets.

B. State of the art

1) CellposeSAM: CellposeSAM (v4.0.4) is a cutting-edge deep learning model that integrates the robustness of the Cellpose framework with the powerful vision transformer backbone of the Segment Anything Model (SAM) developed by Meta AI [29]. Introduced very recently (may 2025), CellposeSAM builds upon Cellpose's original object flow-based segmentation by replacing its convolutional backbone with the transformer-based encoder from SAM, significantly boosting generalization across image modalities [30].

In contrast to prior versions of Cellpose, which required explicit object size input or model retraining for different modalities, CellposeSAM is trained with scale-invariant augmentations and channel shuffling, making it robust to different cell size and channel order. This enables seamless application across diverse datasets without fine-tuning. Additionally, image degradations such as noise, blur, and contrast inversion were applied during training, ensuring that the model remains robust under suboptimal imaging conditions [31].

CellposeSAM is especially relevant in fluorescence microscopy-based colocalization studies that require segmentation of well-defined, object-like structures such as organelles or bright, punctate protein aggregates. Reliable object segmentation can complement intensity-based analyses by enabling object-level colocalization metrics, such as counting the number of aggregates associated with chaperone signal. For this reason, CellposeSAM was included as a candidate segmentation method in this pipeline.

The capacity of CellposeSAM to segment heterogeneous structures under varying signal conditions makes it particularly suited for high-throughput, 3D, multi-channel microscopy data. By producing high-quality segmentation masks from raw image stacks, it enables downstream object-based colocalization calculations such as Manders' coefficients, Dice overlap, and spatial proximity—critical to understanding biological interactions down to the single-cell level [32].

2) nnU-Net: nnU-Net is a deep learning framework built on the U-Net architecture that automatically adapts to a wide range of biomedical image segmentation tasks. What sets nnU-Net apart is its ability to self-configure—adjusting key aspects such as preprocessing steps, network architecture, training schedules, and post-processing routines that I can tune entirely to the dataset. This makes it particularly well-suited for complex biological obtained from fluorescence microscopy, where the structures of interest vary in size, shape, and intensity [33]. In this study, nnU-Net is used to segment mutant Huntingtin (mHTT) aggregates, improving the accuracy of colocalization analysis with CCT1 chaperones.

In contrast to other research methods, nnU-Net is holistic, that is, its automated configuration covers the entire segmentation pipeline. By integrating nnU-Net, this research streamlines aggregate detection, providing clearer insights into CCT1's role in protein aggregation.

3) Napari: Napari [34] is an open-source, Python-based image viewer developed for interactive visualization and annotation of high-dimensional image data. The viewer is designed for scientific use and integrates seamlessly with the Python ecosystem. Its capabilities include real-time GPU-accelerated rendering, support for multi-dimensional image formats such as fluorescence microscopy stacks, and a highly modular architecture that enables deep integration into Python-based analysis workflows [34].

Napari's core architecture is built around the concept of interactive layers. Users can overlay different data types, including raw images, segmentation masks, points, vectors, shapes, and surfaces. Each layer has a customizable set of properties, such as opacity, colormap, and blending mode, allowing users to visually distinguish between modalities and analysis outputs. [34].

Crucial to Napari's utility in this research is its robust support for n-dimensional (nD) data. It is used in this pipeline to build a 3D rendering mode that performs real-time volume rendering [34]. Multi-channel support is native, and users can adjust the display parameters for each fluorescence channel independently. This capability enables clear and interactive visualization of colocalization patterns between mutant huntingtin (mHTT) and the CCT1 chaperone, directly from the same analysis pipeline used to compute colocalization metrics.

Stakeholder Analysis

The development of an automated colocalization analysis pipeline for CCT1 and mutant Huntingtin aggregates must carefully consider the needs and expectations of a diverse set of stakeholders. These include not only the immediate scientific users but also the broader community of potential adopters and contributors. Conducting a structured stakeholder analysis ensures that the tool is both scientifically robust and practically usable, and that ethical, organizational, and sustainability aspects are addressed from the outset.

Table I presents a structured overview of all key stakeholders involved in the project, their roles, levels of influence, interests, and recommended engagement strategies. And Figure 1 maps the stakeholders according to their power and interest.

TABLE I
Stakeholder analysis

Stakeholder	Role	Interest	Infl.	Engagement
Researcher	Thesis lead / dev.	Completion, deployment	High	N/A
Academic Supervisors	Guidance, assessment	Scientific integrity	High	Regular updates
Prinses Máxima Centrum	Data provider, user	Quality colocalization tool	Very High	Clean docs
CureQ	Domain experts	Methodological, biological soundness	High	Validate logic
Tool Developers	Algorithm creators	Tool adoption, validation	Low	Cite properly
Bioimage Analysis Community	Potential users	Usability, reproducibility	Medium	Open-source
Future Students	Users / extenders	Educational, collaborative value	Medium	Maintain extendable code

IV. Requirements

This section outlines the requirements for the colocalization analysis tool developed to study the interaction between mHTT aggregates and the CCT1 chaperone. Requirements are structured into two main categories: Product Requirements and Model Requirements. Each category is framed around legal, ethical, organizational, functional, and technical dimensions, validated through current EU regulatory standards and best practices in scientific computing.

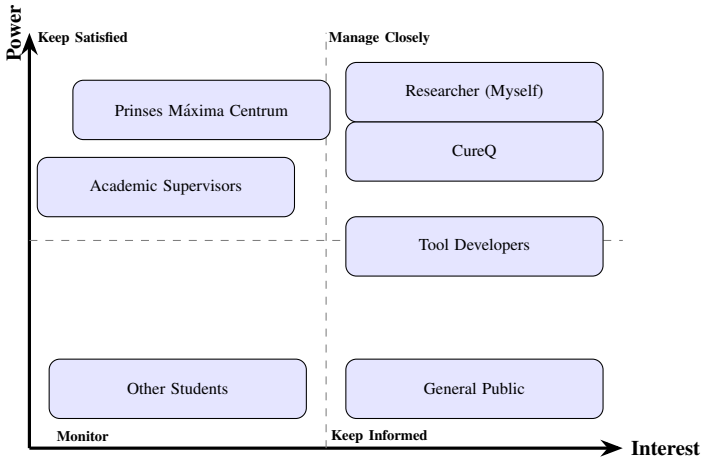


Fig. 1. Stakeholder Power vs. Interest Map.

To prioritize implementation, all requirements has been assigned a term of urgency (eg. Must,Should,Could) using the MoSCoW method [35]:

- **Must**: Critical requirements that are non-negotiable for core functionality.
- **Should**: High-priority features that add substantial value but are not strictly essential.
- **Could**: Useful enhancements that can improve usability or robustness.
- **Won't (for now)**: Out-of-scope requirements that are intentionally excluded from this release.

A. Product Requirements

a) Legal Requirements:

- The tool **must** be used strictly for research use within the EU and not for diagnostic or clinical decision-making. Under Article 2(6) of the EU Artificial Intelligence Act, scientific research applications are exempt from the obligations that are imposed on AI systems under this act [36]
- Since no patient-identifiable information is used, the tool must operate outside the scope of GDPR enforcement under Recital 26 of the General Data Protection Regulation, which clarifies that data is not considered personal when irreversibly anonymized [37].
- All incorporated open-source components **must** be used in accordance with their respective licenses (e.g., MIT, Apache 2.0), and a full list of dependencies and licenses will be maintained on the GitHub page to ensure compliance.
- A research-use-only disclaimer **should** be included in the user documentation and GUI splash screen.

b) Ethical & Societal Requirements:

- The tool **must** not obscure or distort biological interpretations. Outputs must be explainable, traceable, and reproducible by researchers [38].
- Uncertainty in model output (e.g., confidence intervals, segmentation failures) **should** be logged and communicated to users.
- To mitigate disruption to existing laboratory workflows, comprehensive GUI tutorials and short video walkthroughs **should** be provided so that microscopy technicians can transition from manual FIJI workflows to AI-assisted analysis.
- All documentation **must** avoid direct clinical claims; the pipeline is explicitly marked "research-use only" to prevent premature medical deployment or false patient hope.
- The default model **should** run on mid-range consumer GPUs (≤ 10 GB VRAM)

c) Organizational Requirements:

- A version control system **must** be maintained to ensure traceability across iterations.
- User documentation **must** be accessible to non-technical users, with GUI tooltips, tutorials, and example workflows.
- Basic cybersecurity measures **should** follow ISO/IEC 27001 principles, such as secure file handling, up-to-date dependencies, and access control within lab systems [39].

d) Functional Requirements:

- Users **must** be able to load multi-channel z-stack microscopy images, select channels (e.g., CCT1, mHTT), and apply the segmentation models.
- single-image processing **should** have PDF output format.
- **Won't (for now)** — Batch processing of image datasets will not be included in the initial implementation of the tool. The first release focuses on accurate per-image analysis to validate segmentation quality and colocalization metrics in controlled conditions, keeping the human in the loop. Batch mode will be considered for future iterations after core functionality is stabilized and verified.

e) Technical Requirements: In the event of a product failure, it should be designed to fail as gracefully as possible. This includes implementing failure messages during analysis to prevent users from receiving incorrect or misleading outputs [40].

- The tool **should** support .tiff, and .lif file formats, metadata (voxel size, channels) must be retained in analysis.
- Modular architecture **should** enable integration with external segmentation masks (e.g., nuclei).
- Visualization libraries (e.g., Matplotlib, Napari) **should** be integrated for result validation.

B. Model Requirements

a) Legal Requirements:

- Models trained on proprietary data **must** be documented with source metadata and used exclusively within the scope of the Prinses Maxima Centrum research agreement.

b) Ethical & Societal Requirements:

- Model performance **must** be evaluated by experts on microscopy to mitigate biases and ensure biological alignment(these experts will be researchers at the Prinses Máxima Centrum).
- Model generalizability **should** be validated against varied morphological patterns within the lab's dataset.
- Segmentation errors **could** be flagged in the GUI and written to logs.
- In the case of failure, the model (Should) provide a brief model card that refers to documentation on the model's known weaknesses—e.g. reduced accuracy on heavily photobleached images—and outlines practical ways to mitigate those shortcomings(maintained on GitHub).
- Every new version **must** be released alongside the trained weights under an open licence (e.g. CC-BY-NC for weights, MIT for source code) and marked explicitly as "research-use only" to discourage any premature clinical application.

c) Organizational Requirements:

- Training code, model checkpoints, and configuration metadata **must** be versioned and stored in a reproducible format (GitHub).
- Modular inference pipelines **must** allow future reuse and adaptation within a larger continuation project

d) Functional Requirements:

- The model **must** perform segmentation of both mutant Huntingtin (mHTT) aggregates and CCT1 chaperone regions, producing binary masks suitable for spatial overlap analysis.
- The model **must** facilitate specific hyperparameter tuning to cater to differing datasets, for example when channels are ordered differently, this should able to be set by the user.

e) Technical Requirements:

- The model **must** support inference on 3D images with channel resolution normalization in .tiff, and .lif file formats. Retaining metadata (voxel size, channels).
- Any part of the library **should** support GPU-accelerated inference (through CUDA for Nvidia support as well as MPS for mac).
- The pipeline **should** incorporate extensive data augmentation techniques (e.g., rotations, flips, intensity shifts, elastic deformations).

V. Prototype

A. Idea Generation & Value Proposition

The idea for this research emerged from the intersection of two disciplines: advances in computational microscopy and the need to decode protein mechanisms leading to HD. Observations from the Prinses Máxima Centrum on the visual overlap between CCT1 and mHTT aggregates in microscopy

images and subsequent discussion with CureQ sparked an intuitive question: What if we could teach algorithms to see what experts see—and then go further?

This conceptual seed matured into a structured proposal after reflecting on how manual image analysis, while insightful, scales poorly. Meanwhile, some statistical tools like AggreCount offer partial solutions to some of the underlying challenges of the colocalization study and have proven themselves in biomedical segmentation, but none are capable of automating the whole colocalization pipeline. Aggrecount for one quantifies and localizes protein aggregates in single-channel images but does not support multichannel colocalization analysis. it also cannot work with 3D data, making it unable to reap the full benefits of our dataset.

The value proposition of this work lies in its union of usability and precision. Designed to be modular, scriptable, and GUI-accessible. The platform is designed so that experimental biologists can use it easily without needing programming or machine learning expertise, while enabling reproducible, high-throughput analysis.

B. Flow Diagram

The flow diagram shown in Fig.2 illustrates the modular pipeline used to detect and quantify colocalization between mHTT and CCT1 in fluorescence microscopy images. It begins with image preprocessing (including format conversion and z-projection), followed by segmentation of cells, nuclei, and CCT1/mHTT aggregates. Post-segmentation, region mapping aligns CCT1 and mHTT masks to individual cells and—where applicable—nuclear compartments. The final step involves colocalization analysis, using both intensity-based metrics (e.g., Pearson's and Manders' coefficients) and spatial segmentation metrics (e.g., Dice Similarity Coefficient), to assess overlap and proximity at single-cell resolution. Outputs are made accessible through a user-friendly graphical interface, enabling both visual validation and quantitative export.

C. AI Breakdown

The colocalization pipeline pipeline combines deep-learning segmentation with classical statistics in a semi-automated, human-in-the-loop workflow.

Segmentation models *nnU-Net* is trained on the mHTT/CCT1 channels to output four voxel classes—background, mHTT, CCT1, and colocalization. *Cellpose* has been used to provide mHTT masks and optional whole-cell masks that can be used for per-cell aggregation statistics. Both models run fully automatically once the user selects an image, placing the system at *Level-4 autonomy*: the AI proposes a result that the user can accept or override.

Down-stream analytics. From the learned segmentation masks we compute DSC, IoU, PCC and Manders (M1/M2) to describe structural overlap. Pixel-wise enrichment is then quantified with a Mann–Whitney U test and its effect size (Cohen's d), answering whether CCT1 intensity is significantly higher inside aggregates.

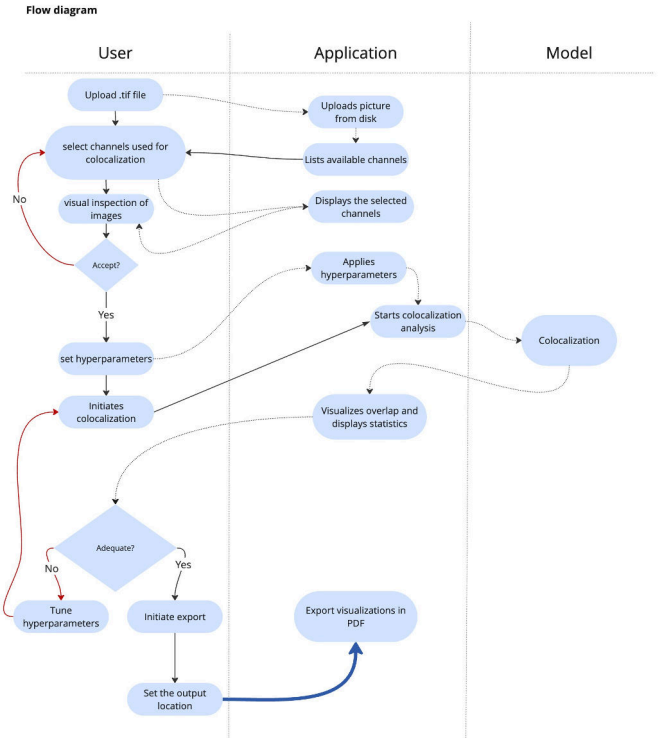


Fig. 2. Flow diagram

Uncertainty sources. Performance depends on (i) segmentation quality—especially for small or diffuse aggregates—and (ii) imaging artefacts that may inflate PCC or Manders when CCT1 is spread out. All numerical outputs are therefore accompanied by visual overlays for expert review.

Level of Automation: The degree of autonomy in this system is classified using the Level of Autonomy (LOA) framework(see: Fig.19), which describes how decision-making is distributed between human and machine [41].According to this framework, autonomy exists on a continuum from Level 1 to Level 10. What each level implies is highlighted in Addendum VIII-D. In the current implementation, the system loads microscopy images and applies default preprocessing and colocalization parameters automatically. These defaults are derived from common fluorescence image analysis conventions and allow users to run the analysis without making manual adjustments. However, users retain the ability to override these parameters if needed as a measure to give control back to the user when automation fails [42].

Classification: LOA Level 4 — The AI suggests one option, and the user decides.

The system executes the default pipeline unless the user intervenes. It does not perform unsupervised parameter optimization, nor does it automatically learn from prior results. Final decisions regarding which results to trust, visualize, or export are left entirely to the researcher. Therefore, the automated pipeline supports human decision-making without replacing it.

Several sources of uncertainty are relevant in the current implementation:

- **Image quality and fluorescence variability:** Background intensity, photobleaching, or spectral bleed-through can distort correlation scores.
- **Thresholding decisions:** Manders' coefficients depend on intensity thresholds to define "positive signal." Errors in thresholding may lead to overestimation or underestimation of colocalization.
- **Lack of spatial context:** Colocalization is measured across entire channels without reference to spatial organization within cells. Proteins may be correlated in intensity but segregated into different compartments.
- **No statistical confidence metrics:** Pearson's and Manders' coefficients are returned as scalar values with no associated uncertainty bounds or distributional estimates.

figure 3 shows the pipeline in a very high-level simple overview.

The colocalization analysis pipeline leverages artificial intelligence primarily for image segmentation, enabling automated, high-resolution analysis of fluorescence microscopy data. Two deep learning models are used at distinct stages:

1. Cellpose – Generalist Cell Segmentation:

Cellpose is a convolutional neural network trained on a broad spectrum of fluorescence microscopy data. In this pipeline, it is applied to segment individual cells, providing spatial context for downstream aggregate mapping. While robust, Cellpose's performance may be supplemented or replaced if a parallel study identifies a more optimized model for the specific dataset used in this research.

2. nnU-Net – Specialized Aggregate and Chaperone Segmentation:

nnU-Net is a self-configuring U-Net-based framework tailored for biomedical image segmentation. It is used here to segment mHTT aggregates and CCT1 regions. Unlike generic U-Nets, nnU-Net automatically adjusts its architecture, preprocessing, and training parameters to biological data.

These AI models automate what would otherwise be labor-intensive and error-prone tasks. However, the system maintains a human-in-the-loop structure: model outputs are verified against biological plausibility and expert-reviewed annotations, ensuring alignment between AI predictions and biological ground truth.

In summary:

- **AI Used For:** Cell and aggregate segmentation (Cellpose, nnU-Net)
- **Traditional Methods For:** Thresholding, morphological filtering, colocalization quantification
- **Level of Automation:** Augmentation (Level 4); human oversight remains essential for validation

This hybrid approach combines the scalability of deep learning with the interpretability of structured analysis, ensuring both robustness and biological relevance.

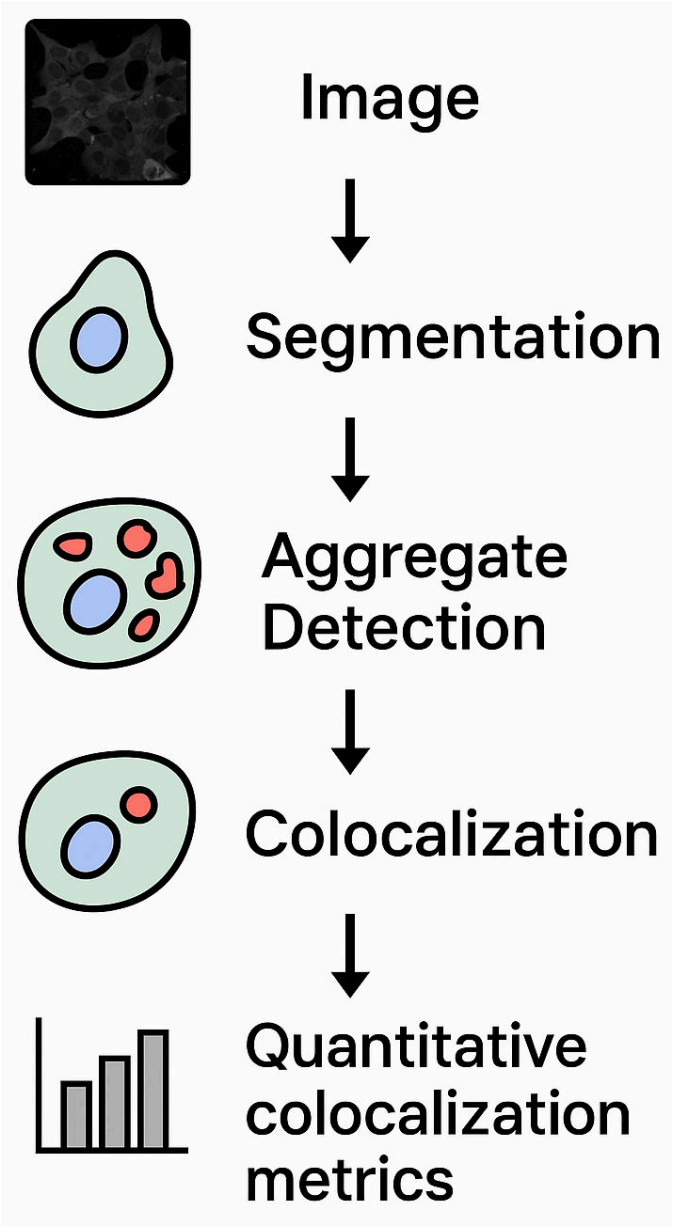


Fig. 3. Segmentation pipeline

D. Users Research

To ensure the colocalization analysis tool aligns with real-world needs, user research has been an integral part of the prototype development process for this project.

The primary end users are researchers at the Prinses Máxima Centrum. Several semi-structured conversations have been held with them throughout the development process to maintain alignment with practical laboratory workflows and analytical requirements. This ongoing collaboration ensures that usability and interpretability remain grounded in the daily reality of biomedical research. Furthermore, the model and accompanying graphical user interface (GUI) has been validated through user testing with this group.

Insights from these conversations were subsequently distilled into an Empathy Map Canvas to systematically capture and communicate the user's goals, frustrations, and practical needs (Fig. 20). This visual summary guided the prioritization of features such as transparency of results, interactive visual validation, and modularity of the analysis pipeline.

In addition, biomedical science students at the Amsterdam University of Applied Sciences have been regularly consulted to provide secondary perspectives on the biological interpretation of intermediate results. These conversations have helped ensure that biological conclusions drawn from the data are robust, comprehensible, and biologically meaningful across audiences with varying expertise.

This close contact with both primary and secondary users has directly influenced prototype design choices, user interface structure, and the prioritization of colocalization metrics, contributing to a more intuitive and useful research tool.

Insights gained from these sessions regarding usability, interpretability of results, and overall user satisfaction will guide refinements for the final deployment.

E. Prototype

To contextualize the value of the proposed automated pipeline, a Supplementary Video has been recorded to demonstrate how this specific colocalization analysis between mutant Huntingtin (mHTT) and the CCT1 chaperone would be performed manually using FIJI [8]. The video walks through the application of colocalization methods. This manual workflow, though informative, demands substantial time investment, manual region selection and inconsistent thresholding. Consequently, it is susceptible to user bias and poses challenges in scaling its application. The proces documented in this video serves as a baseline and showcases the bias-prone nature of the current method.

In contrast, the pipeline developed in this project automates these same analytical steps, improving reproducibility, throughput, and reducing manual error, while still retaining transparency and control over the analysis output.

A Initial prototype was developed in accordance with the requirements and additional insights from the main stakeholders. **This prototype is displayed under addendum VIII-C:**

After validating the prototype, it was adjusted and integrated into an actually working interactive environment within a Jupyter notebook file. This was subsequently used for further testing and finally integrated into a full GUI implementation.

VI. Model

This study employs a hybrid pipeline to quantify the spatial relationship between mutant Huntingtin (mHTT) aggregates and the molecular chaperone CCT1 in 3D fluorescence microscopy images. The approach combines and compares classical image processing, intensity-based enrichment analysis, and deep learning–based segmentation to provide a robust and interpretable assessment of colocalization.

The used images are loaded as multi-channel z-stacks in `.lif` format, with the red and green channels consistently used to represent mHTT and CCT1, respectively. After format conversion and intensity normalization, the mHTT channel was processed using Gaussian smoothing followed by global Otsu thresholding to segment aggregate regions. These binary masks served both as input for model training and as regions of interest for CCT1 enrichment analysis.

Rather than segmenting CCT1 directly, enrichment was assessed by sampling raw CCT1 intensity values within the mHTT mask and comparing them to values in the surrounding background. This strategy avoids the need to create object-based masks the CCT1 channel, since from the analysis it can be concluded that the signal for CCT1 is too dispersed for accurate object-based segmentation.

To complement the intensity-based strategy, object based segmentation was utilized to segment mHTT. This provided voxel-level class predictions across the full 3D volume, enabling additional metrics such as Dice coefficient(III-A2), Intersection over Union(III-A5, Manders' coefficients(III-A1), and Pearson correlation(III-A) to be computed. These segmentation-derived scores offered a structural perspective on colocalization, helping to contextualize the intensity-based enrichment findings.

In some analyses, individual mHTT aggregates were further evaluated for their local CCT1 signal using per-object statistics. This included calculating the signal-to-background ratio, z-score, and percentage of pixels above multiple background-derived thresholds. Aggregates were flagged as colocalized if they exceeded rule-based cutoffs, such as having a z-score above 3 or more than 50% of pixels above a 2σ threshold.

Results were visualized through a combination of raw image overlays, segmentation mask comparisons, intensity boxplots, and scatterplots of aggregate-level measurements. These visualizations, along with structured output tables, supported both quantitative interpretation and qualitative review by biological collaborators.

This pipeline balances simplicity and precision: it leverages the interpretability of intensity-based enrichment while benefiting from the segmentation capabilities of deep learning. Together, these components provide a comprehensive and reproducible framework for assessing proteostasis-related colocalization in complex biological images.

A. Methodology

The core motivation for designing this colocalization pipeline is to automate and relieve researchers of repetitive, error-prone analysis tasks, particularly in high-throughput microscopy workflows. The goal is to achieve a scale advantage—allowing systematic, reproducible quantification of protein colocalization patterns across large datasets that would be unmanageable through manual methods.

1) Data: The data consists of a little over a hundred annotated images from two seperate experiments made with the Leica DMi8 microscope, these images are fluorescence

microscopy images stained with different labels, each represented by a channel (enumerated depending on the specific experiments). There are four channels.

- mHTT-HA (in example red)
- CCT1 (in example green)
- nuclei/nucspot (in example blue)
- A11 – oligomeric species (in example magenta)

Figure 4 illustrates an example of the data. The left channels represent mHTT-HA and CCT1 channels, with an annotated merge of these channels on the right. This example is used throughout the report for consistency and clarity.

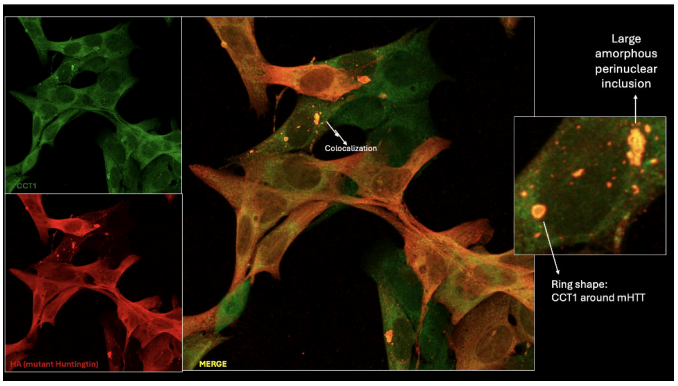


Fig. 4. Ground truth example.

These images are available as .lif file, and contain varying z-values (depending on the experiment and image).

For this research, The mHTT-HA and CCT1 channels are of interest: combining these allows us to explore the question of **whether** mHTT-HA and CCT1 colocalize. In additionally, the third channel (nuclear staining) is used to determine **where** within the cell colocalization occurs (in or outside of nucleus). The final channel in the data is not used, but in the future can be included to provide insight into the molecular form of (less developed) aggregates, that can be used to research **the extent to which colocalization involves oligomeric versus more mature aggregate forms**, thereby offering insight into the aggregation stage at which CCT1 interacts with mHTT.

Within the provided dataset, substantial variability exists across individual images, arising from both microscope settings and biological preparation. Some differences, such as channel order, number of Z-slices, and voxel depth, can be manually adjusted by the user in the pipeline. Other sources, including imaging artifacts (technical imperfections in the image not reflecting true biological structures) and biological variability, are not user-controlled; the pipeline must therefore be robust to these factors or report them transparently.

2) Segmentation Approaches Clarification: In this study, two complementary segmentation approaches are used and compared. The first approach, referred to as **statistical segmentation**, relies on classical image processing techniques, particularly Otsu thresholding and morphological operations, to define mHTT aggregate regions. The second approach, referred to as learned segmentation, uses a deep learning model (nnU-Net based) trained on the dataset to produce voxel-level segmentations of mHTT, CCT1, and colocalization regions. Throughout this report, the terms statistical segmentation and learned segmentation are used to distinguish between these approaches. This distinction is important, as each method offers different strengths and limitations in terms of interpretability, reproducibility, and alignment with biological expectations.

3) Bias: A critical consideration in this study concerns the definition of what exactly constitutes an aggregate. Operational criteria, such as intensity thresholds, minimum area cutoffs, and shape-based filters inevitably introduce bias into the classification process. To segment mutant Huntingtin (mHTT) aggregates, the pipeline employs thresholding to generate an initial binary mask. Following this, morphological operations are applied to eliminate noise by discarding connected regions smaller than a predefined size, and assign unique labels to distinct aggregates. These steps enhance the robustness of aggregate detection by ensuring cleaner boundaries and more accurate analysis. These masks serve as the training labels for the learned segmentation model, thereby transferring any embedded assumptions into the deep learning system.

The accuracy of any model remains bounded by the quality and generalizability of the training data. To mitigate this bias, the segmentation process intentionally aims to including a broad spectrum of detected aggregate shapes ranging from small puncta to dense inclusions captured across varying z-planes and fluorescence intensities. In addition, empirical validation steps were incorporated during preprocessing to identify potential over- or under-segmentation errors. These were assessed based on biological plausibility and visual agreement by a PhD candidate with expert judgment on the dataset. Nevertheless, the inherent challenge of defining aggregates remains a source of uncertainty. Since there is no objective threshold for when a clump of misfolded protein constitutes an aggregate, constant alignment has been carefully safeguarded between the machine learning and neurobiology experts as a bias mitigation technique.

Furthermore, the learned segmentation component of the pipeline was developed and trained on approximately 100 images from two separate experiments. While this represents a relatively large microscopy dataset given the cost and effort involved in generating such data, from a data science standpoint the dataset remains constrained in both size and variability. This introduces significant challenges for model generalization. As a result, the model may not perform reliably on new experiments with different imaging conditions or biological variability. Potential biases inherent in the training data may further limit the model's ability to generalize beyond this specific dataset. Consequently, users of this model have been explicitly informed of these limitations and cautioned against relying on its outputs without appropriate expert validation. This precaution is formalized in the project documentation under the concept of Human-in-the-Loop (HITL), emphasizing that human oversight remains essential when interpreting the model's results.

4) Preprocessing: The preprocessing pipeline is crafted to standardize multi-channel fluorescence microscopy images, facilitating both classical and deep learning-driven analyses. All relevant data is extracted and converted from the `.lif` format to NumPy arrays or NIfTI volumes, utilizing the `AICSImageIO` and `tifffile` libraries. This process ensures the preservation of voxel scale and channel identity. Across all samples, the red and green channels are designated to mHTT and CCT1, respectively. However, users have the flexibility to modify the colormap for visualization purposes, particularly for individuals with color vision deficiencies.

Each channel was normalized to the [0,1] range on a per-sample basis to correct for variations in exposure and intensity scaling. For mHTT segmentation, a light Gaussian blur was applied to reduce noise prior to Otsu thresholding. This was followed by connected component labeling and small-object filtering to generate binary mHTT masks representing individual aggregates. These masks were used to guide CCT1 enrichment analysis (and for the learned segmentation these were used to provide training labels)

5) Structural Colocalization Analysis: The Structural Colocalization analysis is guided using metrics such as the Dice Similarity Coefficient (DSC), Pearson's Correlation Coefficient (PCC), and Manders' Coefficients (M1 and M2), which each attempt to quantify colocalization, but all can be affected by the dispersed nature of the CCT1 signal. DSC measures the spatial overlap between two binary masks, and when CCT1 is found throughout the cytoplasm, the likelihood of incidental overlap with mHTT aggregates increases, resulting in a moderately high Dice score even in the absence of specific targeting. PCC, which evaluates the linear correlation between intensity values across pixels, can give a high score even if the proteins aren't truly interacting—just because both signals happen to be brighter in the same areas regardless of biological relevance, like from uneven lighting. While Manders' coefficients avoid background sensitivity by considering only positive signal overlap, they too might be inflated due to the CCT1 signal being broadly distributed. In particular, Manders M2 may overestimate colocalization when CCT1 is everywhere, and M1 may be misleading when most mHTT pixels overlap with a very spread out CCT1 signal. This is not to say that these metrics are not useful, but they must be interpreted alongside an enrichment test.

To determine if the colocalization is biologically meaningful, the following question must be answered:

> *Is the CCT1 intensity higher inside mHTT aggregates than outside?*

This is a question of enrichment, not just overlap.

To address this question, an enrichment analysis was conducted as follows:

The CCT1 image is divided into two regions:

- Inside mHTT aggregates (as defined by the mHTT mask).
- Outside mHTT aggregates.

The distribution of CCT1 intensities is then compared in these two regions using:

- Mean intensity comparison.
- Mann–Whitney U test to determine if the difference is statistically significant.
- Compute Cohen's d for effect size.

This returns a more objective measure of whether CCT1 is truly enriched at aggregates or just incidentally overlapping.

6) Enrichment Analysis: To determine whether CCT1 fluorescence is enriched at sites of mutant Huntingtin (mHTT) aggregation, voxel-wise enrichment analysis was performed using binary masks to define aggregate regions. Two types of masks were used:

1) **Classical (statistical segmentation) masks:** These were generated through global Otsu thresholding along the various z-slices of the image, followed by morphological operations (e.g., small-object removal and connected component labeling).
2) **ML-based (learned segmentation) masks:** These were produced using the 3D nnU-Net model trained on multi-channel fluorescence z-stacks. The model outputs voxel-wise segmentations across four classes: background, mHTT, CCT1, and colocalization. The mHTT class masks were used to define regions of interest for enrichment analysis. This allowed evaluation of CCT1 signal intensity inside versus outside predicted mHTT aggregates.

In both cases, the *Mann–Whitney U test* was applied to test for statistical significance, and *Cohen's d* was calculated to quantify the effect size of the enrichment. This dual approach enabled comparison between classical and learned aggregate definitions and ensured the robustness of the enrichment results across segmentation strategies.

No binary mask was used for CCT1 during the final preprocessing methodology, as the enrichment strategy operated directly on raw intensity values. Instead, the CCT1 signal was evaluated within and outside of the mHTT mask. Cellpose based whole-cell segmentation masks were generated using the dispersed CCT1 signal(Fig.7) and aligned to facilitate future per-cell aggregation statistics in downstream applications.

For every z-stack the CCT1 intensities *inside* were compared with *outside* the mHTT mask using the non-parametric Mann–Whitney U test. The test avoids normal-distribution assumptions (The fluorescence intensity values are not normally distributed) and is robust to dispersed signals (since it compares relative ranks, not absolute signal levels). The test returns both a p-value and effect size (Cohen's d). Fold-change was computed as the ratio of mean CCT1 intensities in the two regions.

While the pixel-level enrichment results (e.g. Mann–Whitney U test, Cohen's d, Fold-change) provides the most interpretable measure of targeted CCT1 accumulation, structural colocalization metrics(Dice, IoU, Manders) adds robustness and detail.

7) Deployment: To ensure the analysis runs reliably across different setups, the full environment setup and instructions for running the analysis has been made available on the GitHub page.

A graphical user interface (GUI) has been built to allow users to interact with the pipeline without needing to write code. Through the GUI, users are able to load image data, run the analysis, and view the results in a straightforward way. This GUI serves as the primary tool used in user testing and will be used by researchers immediately after deployment.

The outputs of the GUI, including visual overlays and quantitative data, are designed to be easily interpreted and facilitate further biological investigations of colocalization patterns using this pipeline.

B. Results

Please refer to the accompanying GitHub page to see the full results of the various analyses.

1) Visual confirmation: To ensure robustness and biological relevance of the colocalization analysis, three different types of colocalization maps will be compared in this section: 1. Statistical Colocalization Map — generated via classical thresholding and intensity overlap (e.g., Manders, Dice). 2. Machine Learning–Based Map — derived from nnU-Net segmentations trained on the fluorescence microscopy data. 3. Annotated Ground Truth Map — created via expert manual annotation of representative aggregates and CCT1 signal zones. This comparison functions as both an internal validation step, assessing agreement between independent computational approaches, and an external validation step, visually comparing the automated outputs with biologically informed expert interpretation.

An obvious caveat here is that any one sample alone does not serve as proof for an overall generalizable conclusion on performance. Moreover, even the full story of this one image is not being told through the visualization, since the data is analyzed in 3D and the visualizations are based on maximum intensity projections, where all the z-slices are collapsed into a single 2D plane. This compression inevitably obscures depth-specific information, such as whether colocalization occurs consistently across the volume or is confined to certain focal planes, potentially masking spatial nuances. That's why in the GUI there is also a 3D render of the data available and the user can scroll through the z-slices. All the metrics are calculated across every z-slice and then averaged.

To compare the visualized results of the various segmentation methods. A small subset of images, annotated by the researches at the Prinses Maxima Centrum (not included in any training data) has been used as a ground truth. Fig 4 shows an annotated validation image that is used as an example in this report for consistent visual comparison. Performance metrics refer to this same image, based on a printout of the colocalization analysis report generated using standard settings in the GUI, as shown in Fig. 13, Fig. 14, and Fig. 5.

Fig.5 displays the visualized output of the colocalization report from the GUI, run on the annotated example validation image used in the report. Fig.13 and Fig.14 show the used hyperparameters and the results for this analysis respectively. For a more in-depth understanding of the GUI and its functionality, readers are encouraged to run the interface on their own machine (a tutorial video and user manual are provided to guide this process).

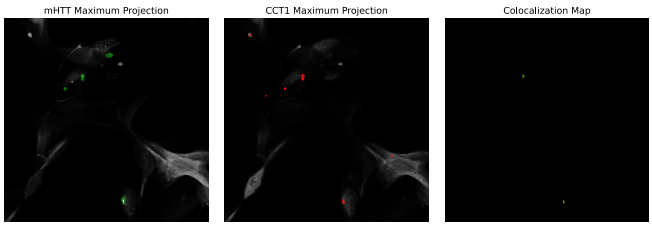


Fig. 5. Colocalization report output (Page 3)

As a further visualization method, Napari [34] has been employed to visualize the colocalization in 3D. This enables investigation of patterns hidden across z-slices that are not readily visible in maximum intensity projections or captured by statistical metrics alone. For example, in Fig. 4, CCT1 is observed to accumulate in a ring-like pattern around a specific mHTT aggregate. In 2D maximum projections of the image, like the one used in Fig. 4, this ring structure is much less apparent. By interactively analyzing the image in 3D, the spatial organization of this ring and its behavior across z-slices becomes clearly visible (Fig. 6). Such ring-shaped CCT1 enrichment patterns were observed around multiple large aggregates in the dataset, supporting the conclusion that CCT1 is not randomly distributed but is spatially organized around mHTT inclusions. This qualitative visualization highlights spatial features of CCT1 enrichment that are not captured by statistical metrics alone, providing additional context for interpreting the colocalization patterns.

2) Comparison of methods:

a) learned segmentation methods: Cellpose-SAM offers a state-of-the-art segmentation approach trained on biological data. However it is not specifically trained on neurons—let

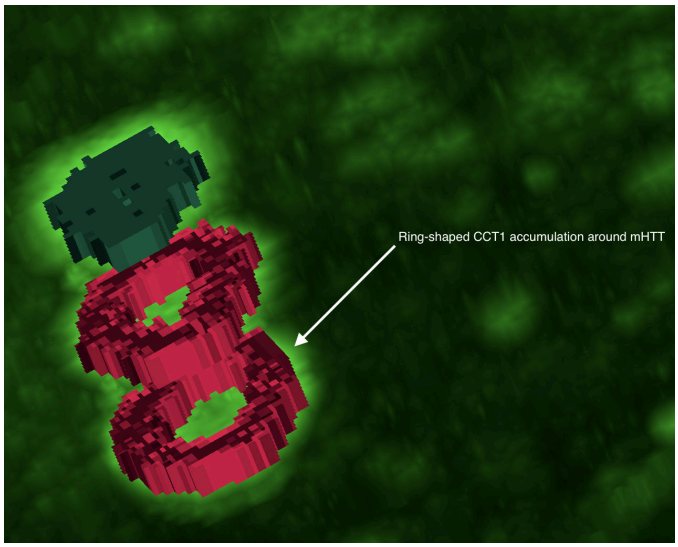


Fig. 6. Zoomed-in view of 3D visualization focused on one colocalization spot

alone on fluorescence microscopy data acquired with the same microscope setup or imaging conditions used in this study. It shows intrinsic and serious difficulties in capturing the underlying patterns required to accurately represent real colocalization patterns. This became most apparent in the validation round performed on the expert-annotated examples, where the CCT1 signal appears too spread out for the learned mask to segment reliably (Fig. 7). In subsequent iterations where learned masks were used exclusively for mHTT detection, its performance on new data was still inferior to statistical segmentation. This too got visually verified when the masks produced in these cases (Fig. 12) were observed to be clearly inaccurate when compared to biological expectations (Fig.4). Across all validation rounds and hyperparameter settings, Cellpose did not outperform the statistically generated colocalization masks. The model appears fundamentally limited by the nature of the training data (differing vastly from our complex data) and the diffuse signal properties of CCT1. With extensive optimization, these performance metrics might improve; however, even then the core issue with this approach is that it proved less flexible. The statistical strategy allows users to tune the analysis to biological expectations by adjusting thresholds and enrichment criteria to reflect the true, diffuse distribution of CCT1, whereas learned segmentation models enforces a fixed segmentation that does not align well with the underlying biological signal.

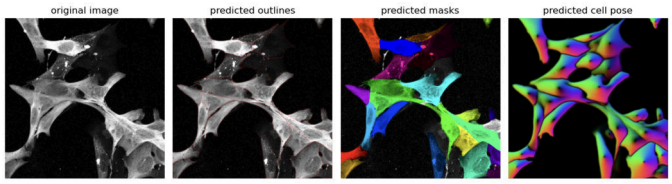


Fig. 7. Learned CCT1 mask (on validation image)

Despite rigorous testing of different nnU-Net configurations, this model was also unable to learn better generalizable patterns from the dataset. The available training data (100 images) is too limited in size and variability to support adequate learning of the complex, multi-faceted colocalization patterns of CCT1 and mHTT. Additionally, due to the highly specific nature of this data, no suitable open-source datasets could be used for effective pretraining. As a result, nnU-Net did not produce segmentations that outperformed statistical approaches.

While, segmentation with nnU-Net does for a full pipeline where the colocalization is judged at the level of individual objects. The use of nnU-Net for structural colocalization analysis comes with significant practical limitations. The model is extremely resource-intensive, requiring high-end GPU hardware, tons of training data and considerable processing time to generate 3D segmentations. In contrast, the statistical analyses can be performed rapidly on standard hardware without training and already provide clear, reproducible answers to the primary biological question of whether CCT1 is enriched at mHTT aggregate sites. Given these trade-offs, for application of this pipeline, the statistical approach offers a superior, more accessible and more scalable solution.

A very extensive pipeline has been constructed to fully automate the implementation of these models. However, since these learned segmentation solutions are not easily human-interpretable and very difficult to guide, they offer little transparency into their decision process, making them near impossible to tune without large amounts of data. This leads to a situation where achieving biological alignment is near impossible. Furthermore, they consistently struggle with key limitations observed throughout this project: The learned segmentation methods are limited by the small size and low variability of the available training set, preventing it from learning robust and transferable representations of the complex colocalization patterns. Together, these limitations render learned segmentation unsuitable for the current colocalization analysis, where statistical methods continue to offer superior robustness and biological interpretability.

Given these limitations, the remainder of this analysis focuses on results obtained through the statistically driven colocalization pipeline, which provided both more reliable performance and clearer biological interpretability across all tested datasets.

b) Statistical segmentation methods: In contrast to the learned segmentation approaches, the statistical segmentation pipeline proved to be aligned with biological expectations in this study. For mHTT aggregate segmentation, the pipeline applies global Otsu thresholding followed by morphological operations, generating binary masks that define aggregate regions. These masks served both as input for the enrichment analysis and as the baseline against which learned segmentations were evaluated. This baseline has not been beaten.

The statistical segmentation approach offers several key advantages for this dataset. First, it is fully automated and parameter-free, eliminating the need for extensive tuning and allowing rapid, reproducible processing of new images that can be manually adjusted. Second, it produces masks that remain consistent with expert biological interpretation in the validation rounds through comparison with expert-annotated examples. Third, it is computationally lightweight and transparent: each processing step is well understood and interpretable, facilitating trust in the resulting colocalization metrics.

In practice, this approach yielded clear and highly significant enrichment results while also supporting reliable structural colocalization metrics.

The implemented multi-stage analysis reveals a consistent picture: the CCT1 chaperone is *not* randomly distributed throughout the cell but is instead strongly and specifically recruited to mutant Huntingtin (mHTT) aggregates. The most important quantitative and qualitative findings that underpin this conclusion are summerized:

3) CCT1 Enrichment: Using an Otsu-derived mask on the mHTT channel as a baseline, CCT1 intensity was on average $\sim$13.69-fold higher within masked aggregates than in the surrounding cytoplasm(Fig. 8). This strong enrichment

indicates that CCT1 is not uniformly distributed but instead accumulates preferentially at mHTT aggregate sites.

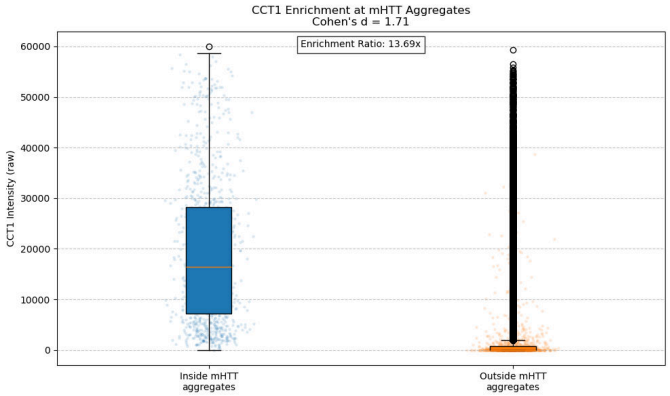


Fig. 8. cct1 enrichment boxplot (using statistical segmentation

Statistical testing confirmed the robustness of this enrichment: the difference in CCT1 intensities inside versus outside aggregates was highly significant (Mann–Whitney $p < 10^{-145}$) and accompanied by a large effect size (Cohen's d 0.90), leaving little doubt that CCT1 is strongly enriched at aggregate sites. The mean CCT1 intensity was 0.89 inside mHTT aggregates compared to 0.84 outside, confirming that CCT1 preferentially accumulates at aggregate sites.

When the same comparison was repeated with learned segmentation masks, the fold change decreased to $\sim$2.5$\times$ (Fig. 9). This drop is expected: ML-based masks include the many low-signal voxels sitting right at the aggregate's edge depleted of CCT1 that hand-tuned thresholding would have excluded, illustrating how the perceived strength of enrichment depends on how "aggregate" is defined.

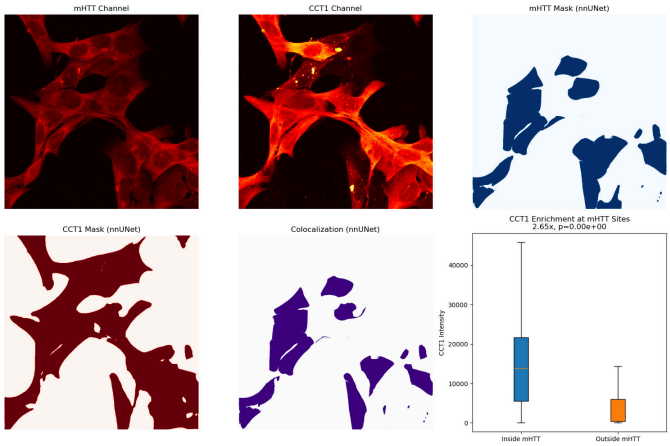


Fig. 9. nnunet analysis visualization

a) Basic overlap metrics.: To complement the enrichment analysis, basic pixel-wise overlap metrics were computed. The mHTT mask covered 4098 pixels, while the CCT1 signal spanned 1360 pixels. Of these, 1206 pixels were shared between both signals, corresponding to 29.43% of the mHTT mask and 88.68% of the CCT1 signal. This high fraction of CCT1 signal residing within mHTT-defined regions further supports the conclusion that CCT1 is preferentially associated with mHTT aggregates, in line with the enrichment and correlation results.

4) Global correlation and overlap: Across entire images, the two fluorescence channels showed a linear relationship (Pearson $r \approx 0.46$),

Manders' coefficients further reinforce this observation: nearly 90% of bright CCT1 signal lies within mHTT regions (M2 = 0.8788), whereas only about 30% of mHTT signal is overlapped by CCT1 (M1 = 0.3060). The pronounced asymmetry indicates that while the vast majority of high-intensity CCT1 signal accumulates at or near aggregates, not all aggregates are equivalently enriched in CCT1, even after accounting for three-dimensional morphology. This pattern is consistent with a targeted recruitment mechanism, rather than a uniform interaction across all aggregates.

Structural overlap between the two signals is clear, with a DSC of 0.4419 and an Intersection over Union (IoU) of 0.2836, further reflecting that CCT1 is selectively associated with aggregates rather than diffusely colocalized throughout the cell.

5) Structure-aware colocalization: Quantitative Otsu's statistical thresholding provides the best balance of simplicity and accuracy (DSC 0.48), vastly outperforming fixed background rules. Although a deep-learning approach (CellposeSAM) could squeeze out a slightly higher overlap (0.53), we know from further visual validation that these masks lack biological alignment. Otsu contributes to a fully automated, parameter-free pipeline that offer a good statistical solution for our downstream colocalization measurements(Fig. 10).

6) Spatial context: Dice overlap between the nucleus mask and both mHTT aggregates (0.0011) and CCT1 signal (0.0002) was negligible, indicating that colocalization occurs predominantly outside the nucleus. Further analysis of nuclear localization is outside the current scope and may be explored in future work. An initial comparison is provided in III.

7) GUI: For deployment, a comprehensive Graphical User Interface (GUI) has been meticulously crafted. Throughout the development process, the GUI underwent iterative refinement, guided by user testing and peer feedback. These successive iterations of the GUI are available on the GitHub page. Alongside the final GUI. There is also a full usage tutorial and User manual available for the GUI, these tools assure a comprehensive user experience.

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Conclusion

This study demonstrates that the segmentation pipeline, as implemented through the graphical user interface (GUI), enables reproducible and robust quantification of CCT1 localisation specifically at mHTT aggregation sites. The biological conclusion has been validated through comparison with expert-annotated ground truth and complementary structural analyses.

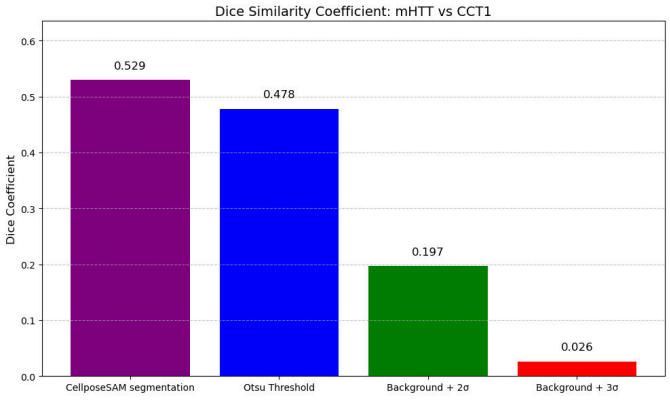


Fig. 10. Dice coefficient barplot for segmentation comparison.

From the perspective of technical, social, and ethical suitability, this project shows that while AI-based learned segmentation offers promising capabilities, it is currently not suitable for reliable and interpretable colocalization analysis in this context. Limitations in training data size, model generalization, and most of all biological alignment, hinder its routine use. Without sufficient data and rigorous validation, reliance on fully automated AI outputs risks introducing misleading conclusions. Such risks are ethically unacceptable in biomedical research.

By contrast, the statistical segmentation pipeline demonstrates robust performance, biological interpretability, and practical usability. It provides a transparent tool for quantifying colocalization in 3D fluorescence microscopy data, enabling researchers to perform reproducible analyses with minimal manual intervention. The pipeline streamlines previously labor-intensive workflows and offers interpretable results that support biological insight and experimental decision-making.

Finally, because the pipeline is modular and model agnostic, it is prepared for future integration of improved models and datasets. This ensures that AI remains an option for enhancing specific components of the pipeline, even though it is not suitable for the current core analysis.

Taken together, these findings provide a clear answer to the research question: the molecular chaperone CCT1 shows significant spatial enrichment at mHTT aggregate sites, and this relationship has been robustly quantified using automated analysis. The developed pipeline effectively overcomes key limitations of manual fluorescence microscopy analysis, streamlines workflows that would otherwise be time-consuming and very prone to bias, and ensures that colocalization can be quantified consistently across experiments. By combining interpretable statistical analysis with human-in-the-loop safeguards, the pipeline offers a transparent and scalable framework for 3D colocalization analysis in complex biological datasets, providing quantitative evidence that supports therapeutic research in this context.

VII. Discussion

A. Ethical Implications

The added value of this pipeline is automation, leading to improved reproducibility, yet this also brings some ethical considerations with it. In this section important ethical implications will be discussed that arise when pattern-recognition is automated.

a) Improved Reproducibility: Manual colocalization (as performed using FIJI) is very sensitive to specific threshold selection (bias), user fatigue, and undocumented last-minute tweaks. By formalising each analytical step in code and exposing every parameter in a graphical interface, the pipeline turns reproducibility into a default. These transparent methods are meant to not only strengthen scientific integrity, but also help reduce the risk that therapeutic hypotheses are pursued on the basis of non-reproducible results. This improved reproducibility forms an ethical foundation for the pipeline's use, but also necessitates careful attention to new risks that arise when automating complex pattern recognition, as discussed in the following sections.

b) Over-reliance on Model Output.: High DSC(III-A2) or IoU(III-A5) scores can create a false sense of security. The users may be tempted to accept good performance metrics as ground truth and skip visual inspection in their research. To prevent inaccurate segmentations from leading to misleading biological conclusions, the GUI always shows the users the overlay images before exporting statistics, since final responsibility for biological plausibility remains with the user, not the model.

c) Bias and Representativeness: The models used tend to inherit the biases encoded in their training data and from the design decisions made during model development. (nnU-Net, for instance, was fine-tuned on the fluorescence microscopy images acquired under consistent exposure and staining conditions.) If those conditions shift(e.g. weaker fluorescence signal, different cell types). Aggregates may be under/oversegmented and CCT1 enrichment wrongfully estimated. A standing invitation therefore exists for stratified validation: performance should be re-evaluated whenever the imaging protocol or sample biology changes in a meaningful way.

d) Opacity and Explainability.: When a prediction fails, users need a way to diagnose why. For now, the pipeline does this through four tactics: *(i)* side-by-side overlays, *(ii)* per-object enrichment scores, *(iii)* mandatory logging of model and code versions, and *(iv)* ML-based and simple statistical approaches for every step that act as a reality check (to prevent having a solution and looking for a problem from that perspective). Future releases can explore importance maps to make misclassifications more intelligible.

e) Shift in Expertise and Accountability.: There are some inherent dangers to the fact that this project lies at the intersection between disciplines. Researchers might obtain a 'good' segmentation without understanding the imaging artifacts that misled the model. Conversely, data scientists may

lack the biological intuition to spot improbable phenotypes. In the development of this process, this has been mitigated by maintaining close collaboration between developers and domain experts. Moving forwards, this alignment process remains an ethical necessity.

f) Data Governance: Although the dataset contains no personal information, fluorescence images still represent a costly experimental asset owned by the Prinses Máxima Centrum. Storing as much information as possible alongside every result file creates a verifiable audit trail while honouring FAIR data principles. [38] Public release of the code base further allows external scrutiny and reuse, therefore the initial implementation (or at least the ML trained part of the implementation) is strictly for internal usage.

g) Potential for Downstream Misuse.: The pipeline is explicitly labelled *"research-use only"*. If repurposed for clinical decision making, it would leave the protective confines of the EU AI Act's research exemption and trigger stricter regulatory duties. Clear documentation and the absence of medical claims in the interface serve as soft barriers against such drift.

Table II summarizes key ethical concerns and how the current implementation addresses them.

TABLE II
KEY ETHICAL CONCERNS AND MITIGATION IN THIS WORK.

Concern	Why it matters	Mitigation
Over-reliance	Inaccurate masks distort colocalization	Mandatory overlay review; numeric output locked until review
Bias amplification	Faint aggregates underrepresented; enrichment skewed	Cross-setting benchmarking; optional nnU-Net fine-tuning
Opacity	Deep models obscure error tracing	Version logging; uncertainty flags; future XAI integration
Downstream misuse	Tool may be misused as diagnostic without validation	Research-use disclaimer; no diagnostic claims in GUI

B. Looking Forward.

As this pipeline matures and is integrated into further applications, three main priorities stand out:

1) To robustly tie biological conclusions to this analysis, it is essential to validate the observed enrichment patterns on the examples in additional experimental contexts and at larger scale. This will be needed to prove that the observed CCT1–mHTT association generalizes at scale.
2) Integration into potential future models and datasets, leading to improved mask generation (this has already been streamlined in the current learned segmentation pipeline).
3) Maintaining a full model card that details all future training data composition and known failure modes.
4) Implementing automated monitoring that warns users when incoming images deviate materially from the data on which the model was last validated, ensuring a graceful failure and handoff [40]. Since even within the provided experiments there was little consistency in the images(eg.channel order, voxel depth and number of Z-slices). Any of these attributes embedded within the images should be taken into account.

a) Per-cell Analysis and nuclei Detection.: While this project focused primarily on quantifying colocalization at the aggregate level, the pipeline was designed with per-cell analysis and nuclei detection as an integral component of its architecture. The results of some analysis on the nuclei masks are presented in the addenda. Additionally, preliminary testing has demonstrated that the current learned segmentation pipeline can accurately segment nuclei from these images (Fig. 11). However, this implementation has been suspended pending the development of a more accurate nucleus segmentation pipeline by a parallel study.

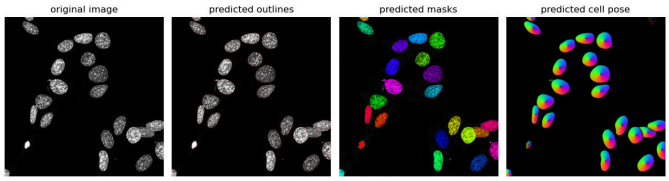


Fig. 11. Learned segmentation mask for nucleus

The current implementation includes support for nuclei and cell segmentation through the integration of Cellpose, and the alignment of these masks with aggregate masks has been accounted for in each step of the pipeline design. However, this component has not yet been fully optimized or systematically validated, as the main focus of this study was on establishing robust and reproducible aggregate-level colocalization analysis. Future work can readily extend the pipeline to support refined per-cell quantification.

b) Future adaptation of the learned segmentation pipeline: While the learned segmentation methods evaluated in this study did not yield sufficient performance for reliable colocalization analysis under the current data constraints, the pipeline itself was intentionally designed with future extensibility in mind. The segmentation component is implemented in a modular and model-agnostic manner, allowing researchers to replace or augment the nnU-Net-based segmentation with different segmentation models as needed.

It is important to note that the primary limitation observed in this project was the size and variability of the available training data, not the pipeline architecture itself. With access to larger and more diverse microscopy datasets, future researchers can experiment with more powerful models and fully leverage the learned segmentation path of the pipeline.

In this way, the pipeline is prepared to evolve alongside future methodological advances, while ensuring that robust and biologically meaningful colocalization analysis remains the central priority.

Future work can also aim to extend the current analysis by integrating an **Oligomeric Species Analysis:** Using the fourth fluorescence channel (A11), which marks oligomeric protein species, future analysis could assess whether CCT1

preferentially colocalizes with oligomeric (rather than fibrillar) aggregates. This may reveal whether CCT1 interacts early in the aggregation process or binds predominantly to mature structures, offering further insight into its potential role in proteostasis and therapeutic targeting.

VIII. Addenda

A. Learned Segmentation results

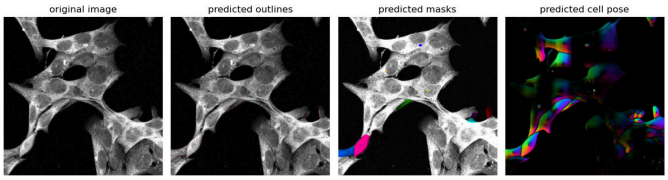


Fig. 12. Learned mHTT mask (on validation image)

For the learned segmentation approach alongside Cellpose, nnU-Net was used, a self-configuring deep learning architecture tailored for biomedical image segmentation. The model was trained on 3D volumes composed of the mHTT and CCT1 channels and produced voxel-level predictions for four classes: background (0), mHTT (1), CCT1 (2), and colocalization (3). Training labels were derived from thresholded mHTT masks, that were created by statistical means (These perform better then the CellposeSAM masks) and curated annotations for CCT1 and overlapping regions. For nnU-Net training, an 80%,20% train/test split was performed. The final segmentation and colocalization metrics are based on an independent test set. nnU-Net was selected for its automated configuration, streamlines the pipeline and reduces the need for manual tuning of (hyper)parameters. This made it highly scalable and suitable for processing z-stacks across large datasets.

The GUI's printout when executed during the experimentation is presented in Fig. 14. In contrast, the learned segmentation implementation achieved the following results:

- CCT1 inside mHTT aggregates: mean = 52,008.27; median = 54,160.50
- CCT1 outside mHTT aggregates: mean = 7,299.31; median = 1,798.00
- Enrichment ratio (inside/outside): 7.13×
- Mann-Whitney U test: p-value = 0.000e+00
- Effect size (Cohen's d): 5.20
- Interpretation: CCT1 is strongly enriched at mHTT aggregate sites (large effect)

This pattern of enrichment is also clearly visualized in the corresponding boxplot (Fig. 9), which underscores the magnitude of CCT1 accumulation within aggregates.

Spatial Overlap with Nucleus

For completeness, Table III reports Dice similarity coefficients between the nucleus mask and both mHTT and CCT1 masks, using both statistical and Cellpose-derived segmentations. These comparisons serve as some initial results hinting that observed colocalization might occur primarily outside the nucleus. Detailed analysis of nuclear localization is beyond the current scope and may warrant further investigation.

Comparison	DSC
Nucleus vs. mHTT (statistical mask)	0.0011
Nucleus vs. CCT1 (statistical mask)	0.0002
Nucleus vs. mHTT (Cellpose-derived whole-cell mask)	0.4559
Nucleus vs. CCT1 (Cellpose-derived whole-cell mask)	0.4499

TABLE III
DICE SIMILARITY BETWEEN NUCLEUS AND MHTT/CCT1 MASKS, USING STATISTICAL AND CELLPOSE-DERIVED SEGMENTATIONS.

B. GUI output on validation image

The Figures below display the output of the colocalization report from the GUI, run on the annotated example image used in the report. Fig. 13 showing the used hyperparameters, Fig. 14 the results and Fig. 5 the visualizations

Colocalization Analysis Report

Generated on: 2025-06-10 12:05:47

:h: 20240212_CKR_E10B_mHtt-HAQ97doxy48h_BafA1_HA-STAR580_CCCT1-STAR635P_A11-STAR460L_nucspotlive488_3.lif

:h: 20240212_CKR_E10B_mHtt-HAQ97doxy48h_BafA1_HA-STAR580_CCCT1-STAR635P_A11-STAR460L_nucspotlive488_3.lif

Analysis Mode: Maximum Projection

Analysis Parameters:

mHTT threshold factor: 4.0

CCT1 threshold factor: 4.5

Gaussian sigma: 3.5

Min object size: 20 pixels

mHTT color: green

CCT1 color: red

Fig. 13. Colocalization report output (Page 1)

C. Initial Figma prototype

The user starts off by selecting the application (Fig. 15), then a screen appears where the user can select the files to perform the analysis on (Fig. 16). After selecting a file, the user can select the channels they want to use for the colocalization (Fig.17). They can also slide through the z-slices of these channels, as every slice can be analyzed individually. If they like it, they can accept, which leads them to the next screen (Fig. 18) where they can change the hyperparameters to tune the analysis. run the colocalization when they are ready. The user is then asked where they want to save the results. Additionally the user can display a 3D visualization for inspection.

D. LOA framework

Applying Artificial Intelligence to this task is done to automate and relieve researchers of certain tasks. However, as automation increases, so do the associated risks, including potential over-reliance on AI and reduced human oversight. This trade-off is captured by the *Level of Automation (LOA)* framework, which describes how AI systems distribute control

Colocalization Metrics Summary

Basic Metrics:

mHTT area: 4098 pixels

CCT1 area: 1360 pixels

Overlap area: 1206 pixels

Overlap percentage: 29.43% of mHTT, 88.68% of CCT1

1. Pearson's Correlation Coefficient:

r = 0.4553 (p-value: 9.96231336e-217)

2. Manders' Coefficients:

M1 (fraction of mHTT overlapping with CCT1) = 0.3060

M2 (fraction of CCT1 overlapping with mHTT) = 0.8788

3. Dice Similarity Coefficient:

Dice = 0.4419

4. Mann-Whitney U Test:

U = 857398.50

p-value = 1.80846847e-145

5. Cohen's d (Effect Size):

d = 0.8962

Mean intensity in overlap: 0.89

Mean intensity outside overlap: 0.84

6. Intersection over Union (IoU):

IoU = 0.2836

Report saved to: colocalization_reports/colocalization_report_20250610_120547.pdf

Fig. 14. Colocalization report output (Page 2)

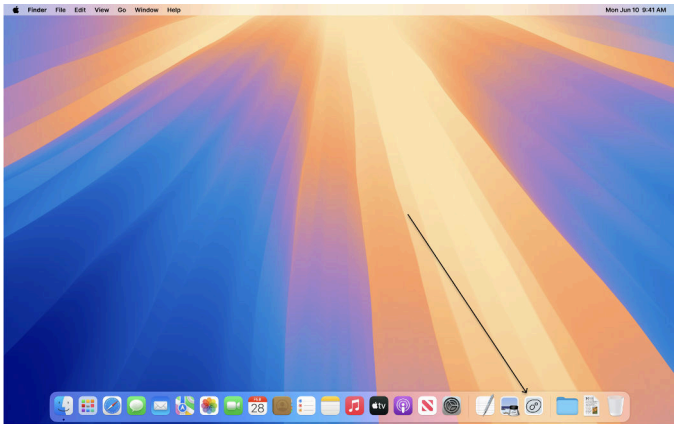


Fig. 15. prototype (step 1)

and decision-making between human and machine [41]. According to this framework, autonomy exists on a continuum from Level 1 to Level 10, where:

- **Level 1** involves no AI assistance—humans make all decisions and actions.
- **Level 10** represents full autonomy—AI acts independently and ignores the human altogether.
- **Levels 2–5** are considered augmentation, where AI supports or enhances human actions.

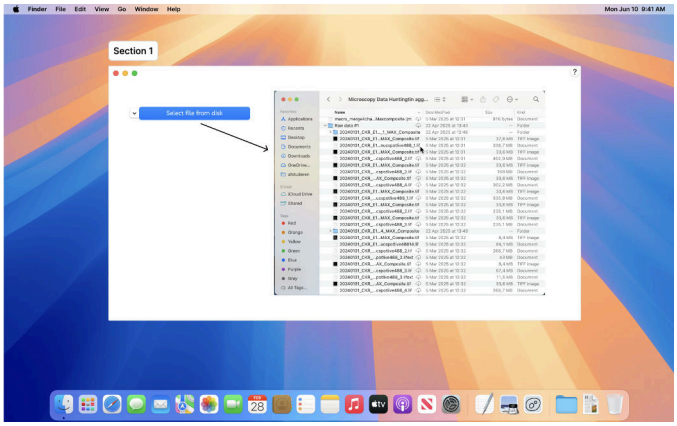


Fig. 16. prototype (step 2)

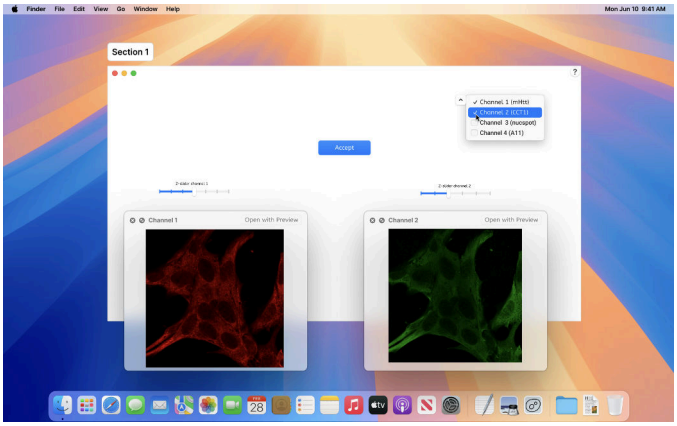


Fig. 17. prototype (step 3)

- **Levels 6–10** are automation, where AI performs the task, though a human is still in the loop at every level below 10.

E. Empathy map

Fig.20 presents an Empathy Map created to summarize the needs, goals, and constraints of the primary user of the colocalization analysis tool, based on discussions with the end user (Carolina, PhD candidate at Prinses Máxima Centrum). This map informed the design and evaluation priorities of the prototype.

F. Acknowledgments

I would like to express my sincere gratitude to the researchers at the Prinses Máxima Centrum, particularly Carolina, for their invaluable insights and guidance throughout this project. I also thank my academic supervisors at the Amsterdam University of Applied Sciences for their continuous support and constructive feedback. Finally, I would like to thank the CureQ consortium for entrusting me with this fascinating subject and for their continued daily support throughout the development of this work.

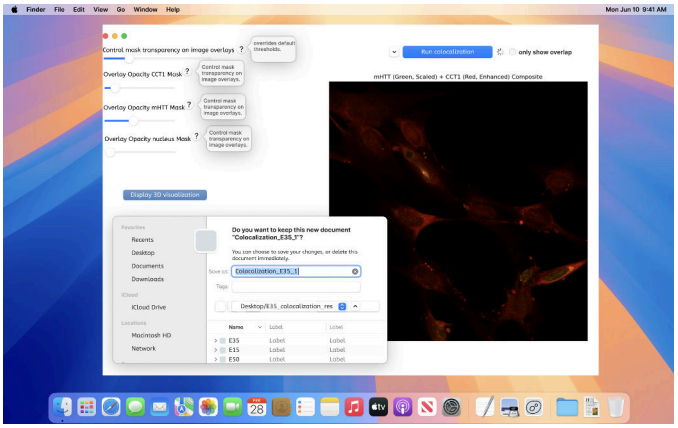


Fig. 18. prototype (step 4)

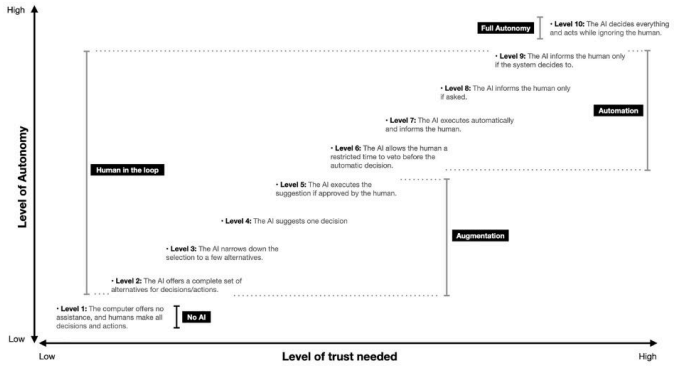


Fig. 19. levels of automation [41]

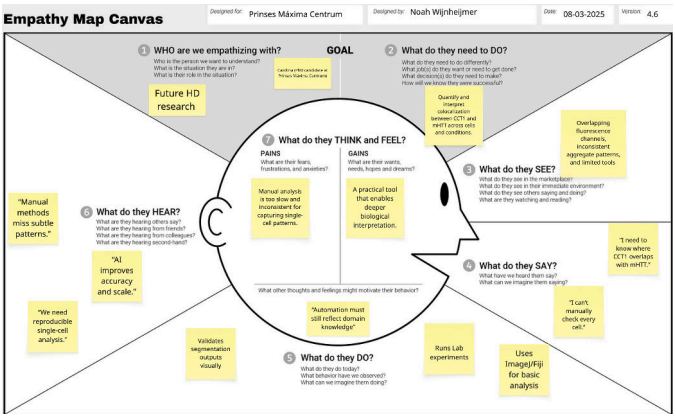


Fig. 20. Empathy Map Canvas summarizing insights from user conversations, highlighting core needs and design priorities for the colocalization analysis tool.

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